Representation of Data

ANSWER KEY

Stem-and-leaf diagrams

1 The data set $23, 25, 31, 34, 34, 38, 42, 45$ is to be shown as a stem-and-leaf diagram with stems $2, 3, 4$. Write out the diagram, and use it to find the median.

Stem-and-leaf: $2 \mid 3\,5$; $3 \mid 1\,4\,4\,8$; $4 \mid 2\,5$ (key: $2 \mid 3 = 23$). With 8 values, the median is the average of the 4th and 5th values: 34 and 34, so median $= 34$.

2 The data set $17, 19, 22, 26, 26, 29, 33, 37, 41$ is to be shown as a stem-and-leaf diagram with stems $1, 2, 3, 4$. Write out the diagram, and use it to find the range.

Stem-and-leaf: $1 \mid 7\,9$; $2 \mid 2\,6\,6\,9$; $3 \mid 3\,7$; $4 \mid 1$ (key: $1 \mid 7 = 17$). Range $= 41 - 17 = 24$.

Histograms and frequency density

3 A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \dfrac{\text{frequency}}{\text{class width}} = \dfrac{25}{10} = 2.5$.

4 A class interval of width 4 contains 18 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \dfrac{\text{frequency}}{\text{class width}} = \dfrac{18}{4} = 4.5$.

5 A class interval of width 5 contains 30 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \dfrac{\text{frequency}}{\text{class width}} = \dfrac{30}{5} = 6$.

Reading grouped frequency data from a bar chart

6 The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.

Total students $= 12 + 28 + 15 + 5 = 60$. The modal group (highest bar) is $3\text{-}5$ books, with 28 students.

7 Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

Students reading 6 or more books: $15 + 5 = 20$. Percent of the total 60 students: $\dfrac{20}{60} \times 100\% \approx 33.3\%$.

Measures of central tendency for grouped data

8 A grouped data set has class midpoints and frequencies: 5 (freq 4), 15 (freq 8), 25 (freq 6), 35 (freq 2). Estimate the mean.

$$\bar{x} = \frac{5(4) + 15(8) + 25(6) + 35(2)}{4 + 8 + 6 + 2} = \frac{20 + 120 + 150 + 70}{20} = \frac{360}{20} = 18.$$

9 A grouped data set has class midpoints and frequencies: 10 (freq 3), 20 (freq 9), 30 (freq 5), 40 (freq 3). Estimate the mean.

$$\bar{x} = \frac{10(3) + 20(9) + 30(5) + 40(3)}{3 + 9 + 5 + 3} = \frac{30 + 180 + 150 + 120}{20} = \frac{480}{20} = 24.$$

Box-and-whisker plots

10 A data set has minimum $= 8$, $Q_1 = 15$, median $= 22$, $Q_3 = 30$, and maximum $= 40$. State the interquartile range, and determine whether a value of 45 would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 30 - 15 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 30 + 22.5 = 52.5$. Since $45 < 52.5$, the value 45 would NOT be considered an outlier.

11 A data set has minimum $= 5$, $Q_1 = 12$, median $= 18$, $Q_3 = 27$, and maximum $= 50$. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 27 - 12 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 27 + 22.5 = 49.5$. Since the maximum value $50 > 49.5$, it WOULD be considered an outlier.

12 A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 32 - 20 = 12$. $1.5 \times IQR = 1.5 \times 12 = 18$. Lower outlier boundary $= 20 - 18 = 2$.

Skewness from summary statistics

13 A data set has mean $= 42$, median $= 38$, and mode $= 35$. Describe the skewness of the distribution.

Since mean $>$ median $>$ mode, the distribution is positively skewed (skewed to the right), with a longer tail of higher values pulling the mean above the median.

14 A data set has mean $= 28$, median $= 32$, and mode $= 35$. Describe the skewness of the distribution.

Since mean $<$ median $<$ mode, the distribution is negatively skewed (skewed to the left), with a longer tail of lower values pulling the mean below the median.

Comparing distributions using summary measures

15 Two classes take the same test. Class A has mean 68 and standard deviation 5. Class B has mean 71 and standard deviation 12. Compare the two classes' performance and consistency.

Class B has a higher mean score (71 vs. 68), so on average Class B performed better. However, Class A has a much smaller standard deviation (5 vs. 12), meaning Class A's scores were more consistent/less spread out than Class B's.

16 Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume 500 mL and standard deviation 2 mL. Machine Y has mean fill volume 498 mL and standard deviation 8 mL. Compare the two machines' accuracy and consistency.

Machine X's mean (500 mL) matches the target exactly, while Machine Y's mean (498 mL) is 2 mL below target. Machine X also has a much smaller standard deviation (2 mL vs. 8 mL), meaning its fill volumes are far more consistent. Machine X is therefore both more accurate and more consistent than Machine Y.

A well-designed word problem: analyzing delivery times

17 This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean $=34$ minutes, standard deviation $=6$ minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.

(a) Threshold $=34+2(6)=34+12=46$ minutes. (b) 44 minutes is $\frac{44-34}{6}=\frac{10}{6}\approx 1.67$ standard deviations above the mean. Since $1.67<2$, this delivery is NOT classified as late.